Extreme Value Analysis of Summer Heat and Dry Events

1 Introduction

Extreme climate events such as heat waves and prolonged dry periods can have significant impacts on ecosystems, agriculture, and human health. Understanding their statistical behavior is therefore an important problem in environmental statistics.

Extreme Value Theory (EVT) provides statistical tools specifically designed to model rare events occurring in the tails of distributions. Two main EVT frameworks are commonly used: the Block Maxima approach based on the Generalized Extreme Value (GEV) distribution, and the Peaks Over Threshold (POT) approach using the Generalized Pareto Distribution (GPD).

In addition to modeling the magnitude of extremes, point process models such as Poisson and non-homogeneous Poisson processes can be used to study the temporal occurrence of extreme events.

The objective of this study is to analyze extreme summer temperatures and drought-related events using EVT methods.

2 Data Description

The dataset consists of daily meteorological observations including temperature and precipitation over several years. From the raw data we construct daily maximum temperature, daily precipitation, and a dry-day indicator based on a precipitation threshold.

Since extreme temperatures mainly occur in summer, the analysis focuses on June–August (JJA).

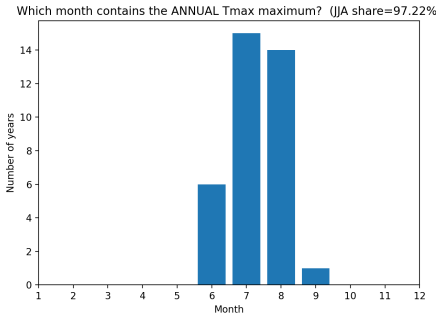


Figure 1: Months in which annual maximum temperatures occur, showing that most extremes happen in summer.

3 Exploratory Analysis

Before applying EVT models, we examine the empirical distribution of temperature observations.

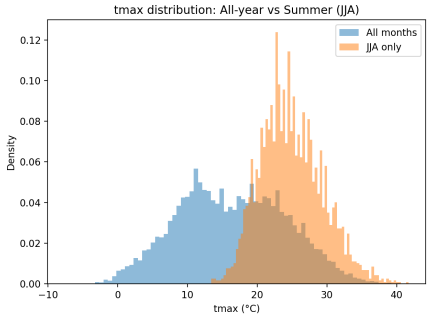


Figure 2: Comparison between the temperature distribution for the entire year and the summer season. The summer distribution shows a heavier upper tail.

4 Block Maxima Approach

The block maxima method considers the maximum observation within each block. Here, each block corresponds to a summer season (June–August), and the maximum daily temperature is extracted for each year.

According to Extreme Value Theory, block maxima follow the Generalized Extreme Value (GEV) distribution. The model parameters are estimated by maximum likelihood, and diagnostic plots are used to assess the model fit.

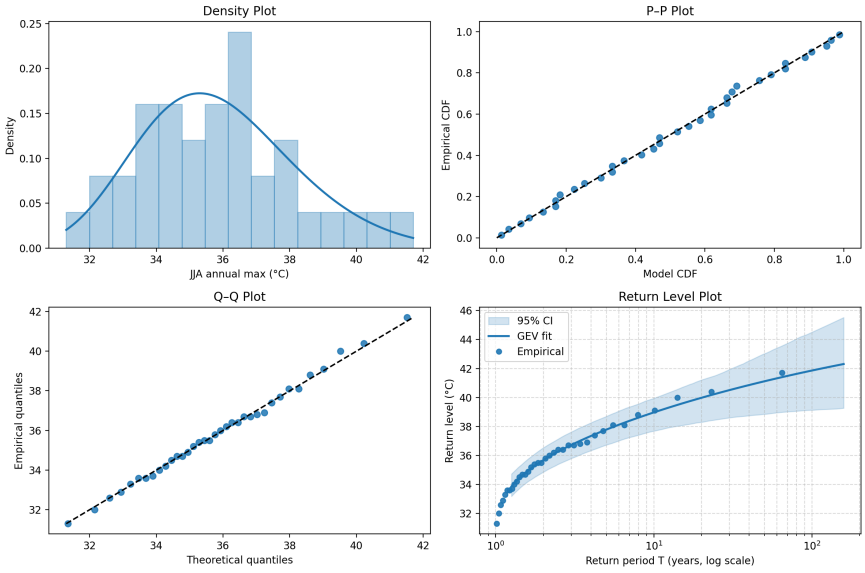


Figure 3: Diagnostic plots for the fitted GEV model: density plot, probability plot (PP plot), quantile plot (QQ plot), and return level plot.

The density plot shows that the empirical distribution of the block maxima is well approximated by the fitted GEV density.

The PP and QQ plots indicate a good agreement between empirical and theoretical values, with most points lying close to the diagonal line, suggesting that the GEV model adequately captures the distribution and the upper-tail behavior.

The return level plot interprets extreme temperatures in terms of recurrence periods. The fitted model shows increasing return levels for larger return periods; for example, temperatures around 40°C correspond roughly to return periods of about 10–15 years, while values near 42°C are associated with much longer periods.

The empirical estimates lie close to the fitted curve, and the 95% confidence interval widens for larger return periods, reflecting increasing uncertainty due to extrapolation.

5 Peaks Over Threshold Analysis

To complement the block maxima approach, a Peaks Over Threshold (POT) analysis was performed on summer daily maximum temperatures. For a high threshold u, the exceedances

$$Y = X - u \mid X > u$$

are modeled using the Generalized Pareto Distribution (GPD).

The POT method uses all exceedances above the threshold instead of only one maximum per year, allowing a more efficient use of extreme observations.

5.1 Declustering

Extreme temperatures often occur in heatwaves, meaning consecutive exceedances may be temporally dependent. To reduce this dependence, a run declustering procedure was applied.

Exceedances separated by less than 3 days were grouped into the same cluster, and only the cluster maximum was retained. This produces approximately independent extreme events for the POT model.

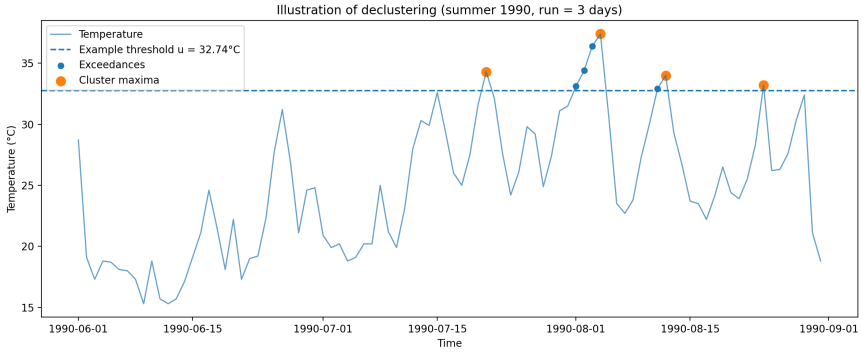


Figure 4: Example of threshold exceedances and declustering.

5.2 Threshold selection

The threshold was chosen using the mean excess plot and parameter stability plots. For a valid GPD approximation, the mean excess function

$$e(u) = \mathbb{E}[X - u \mid X > u]$$

should display an approximately linear behavior above the selected threshold. Additionally, the estimated GPD parameters should remain stable as the threshold varies.

Based on these diagnostics, the threshold

$$u^* = 34.8°C$$

was selected, providing a good compromise between model validity and sample size.

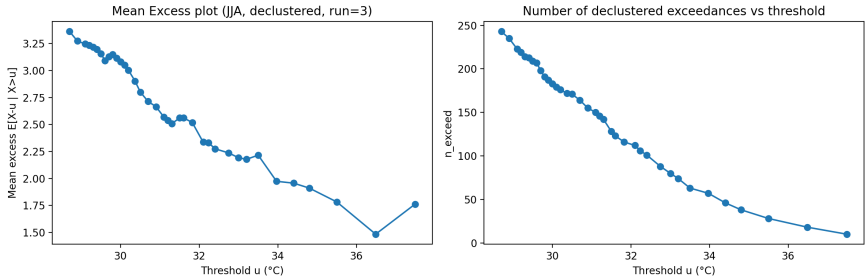


Figure 5: Threshold diagnostics: mean excess and number of declustered exceedances

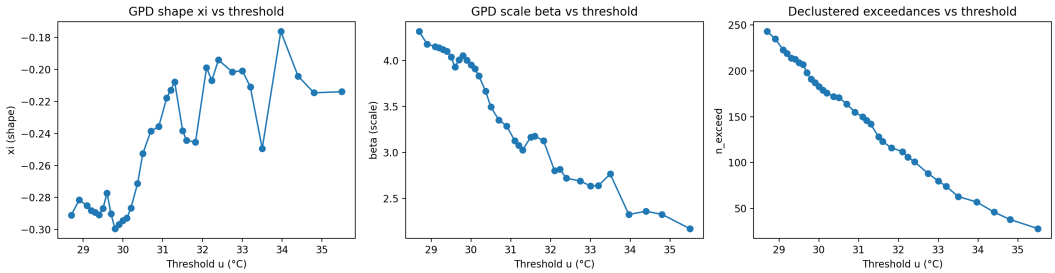


Figure 6: Stability of GPD parameters across thresholds.

5.3 GPD fit and diagnostics

After declustering, the exceedances above u^* were fitted with a Generalized Pareto Distribution (GPD) using maximum likelihood estimation.

The estimated parameters are

$$\xi = -0.2146, \qquad \beta = 2.3260.$$

The negative value of the shape parameter ξ suggests a short-tailed behavior for the distribution of extreme temperatures. In extreme value theory, $\xi < 0$ corresponds to the

Weibull domain of attraction, which implies the existence of a finite upper bound for the distribution of extremes.

Model adequacy was evaluated using several diagnostic plots shown in Figure 7, including the density plot, the probability plot (PP plot), the quantile–quantile plot (QQ plot), and the return level plot.

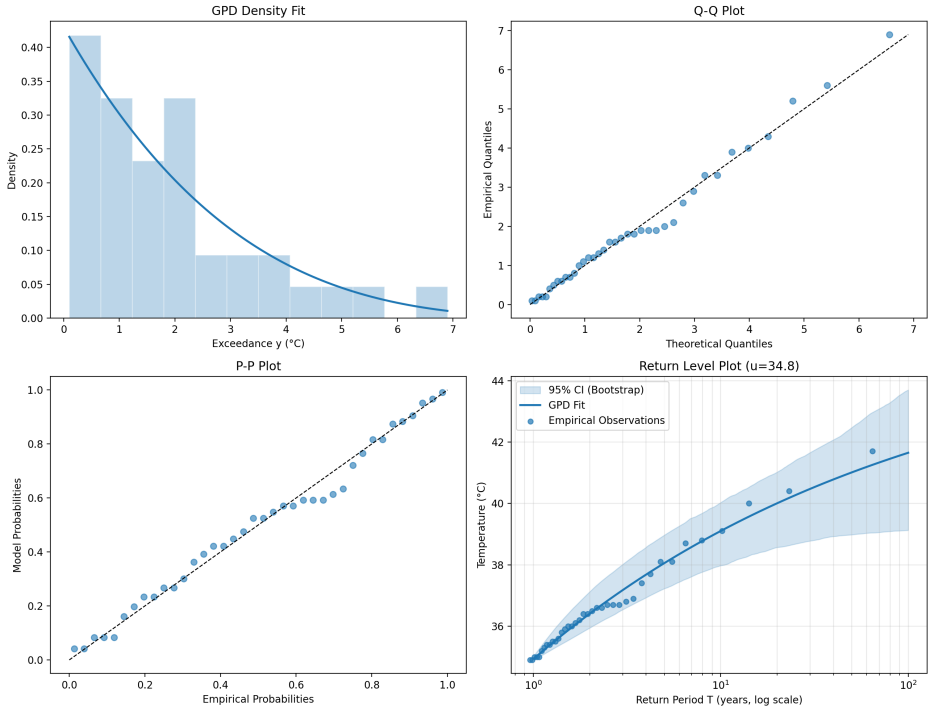


Figure 7: Diagnostic plots for the fitted GPD model: density plot, PP plot, QQ plot and return level plot.

The density plot shows that the empirical distribution of the exceedances is reasonably well approximated by the fitted GPD density.

The QQ plot indicates that the empirical and theoretical quantiles are closely aligned, suggesting that the fitted GPD model reproduces the distribution of exceedances, including the upper tail.

The PP plot also shows points close to the diagonal reference line, confirming that the fitted model provides a satisfactory fit to the cumulative distribution.

The return level plot provides an interpretation of extreme temperatures in terms of recurrence periods. The horizontal axis represents the return period T (in years, shown on a logarithmic scale), while the vertical axis shows the corresponding return level. The fitted GPD curve describes how the magnitude of extreme temperatures increases as the return period becomes larger.

The empirical observations lie close to the fitted curve, indicating that the model captures the relationship between return period and extreme temperature reasonably well. For example, the fitted model suggests that temperatures around 39°C correspond roughly to a return period of about 10 years, while temperatures near 40°C correspond to a return period

of approximately 20 years.

The shaded region represents the 95% bootstrap confidence interval. The interval becomes wider for larger return periods, reflecting the increasing uncertainty when extrapolating beyond the range of the observed data.

5.4 Threshold sensitivity

To assess robustness, the analysis was repeated for several thresholds between $33.5°C$ and $35.5°C$.

The results show similar parameter estimates (ξ around -0.20) and consistent return level estimates (about $39°C$ for 10 years and $40°C$ for 20 years). This indicates that the conclusions of the POT analysis are stable with respect to the threshold choice. Moreover, the consistently negative estimates of the shape parameter across thresholds further support the presence of a bounded upper tail for extreme summer temperatures.

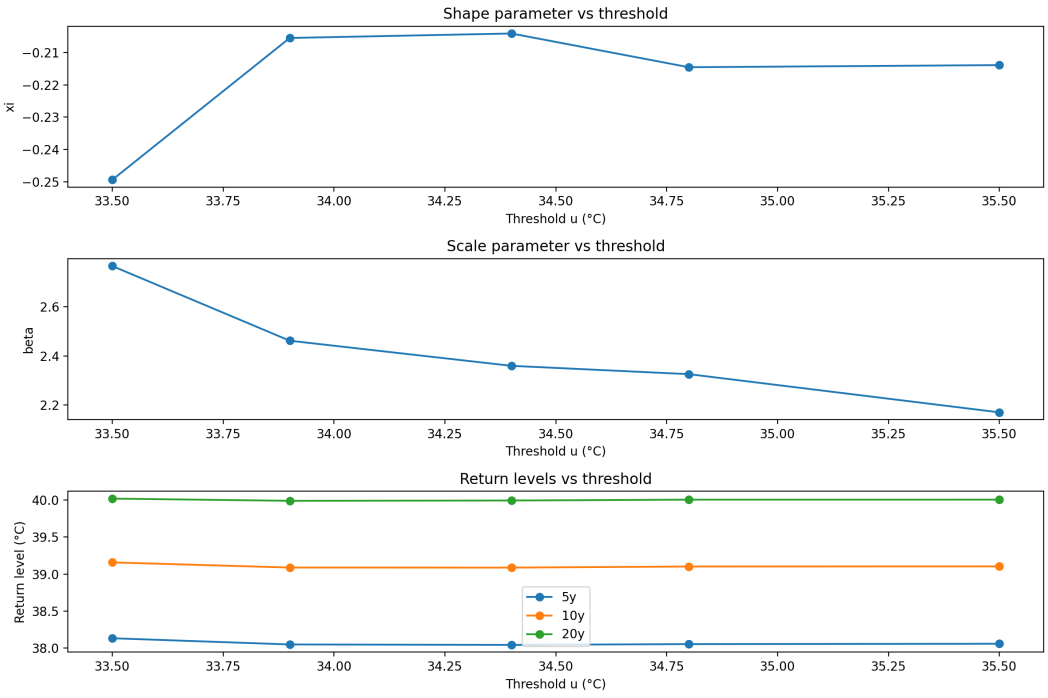


Figure 8: Sensitivity of parameter estimates and return levels to the threshold choice.

To complement the EVT analysis of summer temperature, we now consider the timing of event occurrence.

The GEV and POT models describe the marginal behavior of high temperatures, but they do not explain when drought-related events occur during the summer. To address this, we adopt a point-process perspective.

Several event definitions are possible, including heat-event peaks, dry-spell starts, and compound hot-dry events. In this study, we retain summer dry-spell starts as the main point-process object. This choice is motivated by their clearer event definition and their

larger sample size, whereas heat-event peaks and compound events are much sparser and therefore less suitable for a separate occurrence-time analysis.

Accordingly, the point-process section focuses on the start times of summer dry spells, while the magnitude of heat extremes remains described by the GEV and POT models.

6 Point Process Analysis of Summer Dry-Spell Events

6.1 Event construction

We define a dry day as a day with precipitation below the threshold

$$P_t \leq 0.1.$$

A dry spell is then defined as a sequence of at least three consecutive dry days.

In the point-process analysis, each dry spell is counted as a single event and represented by its starting time. This avoids treating every day within the same spell as a separate occurrence and provides a more meaningful event-based description of drought occurrence.

Restricting attention to the summer season (June–August), we rescale time so that

$$t \in [1, 92]$$

denotes the day of summer. Each event is therefore represented by

$$t_i \in [1, 92],$$

where t_i is the starting day of the i-th summer dry spell.

The resulting dataset can thus be viewed as a collection of seasonal point patterns, one for each year, with the points corresponding to dry-spell start times.

6.2 Poisson point-process model

Let $N(t)$ denote the number of dry-spell events observed up to summer day t. As a first approximation, the occurrence times are modeled as a Poisson point process on the seasonal window $[1, 92]$.

As a baseline, we consider a homogeneous Poisson process with constant intensity $\lambda(t) = \lambda$. To allow for seasonal variation, we also fit a piecewise-constant non-homogeneous Poisson process (NHPP), where $[1, 92]$ is divided into 10-day bins and the intensity is constant within each bin:

$$\lambda(t) = \lambda_b, \qquad t \in \text{bin } b.$$

The model is estimated using Poisson regression on event counts with exposure offsets. For each year y and bin b,

$$Y_{yb} \sim \text{Poisson}(\lambda_b e_{yb}),$$

where Y_{yb} is the number of dry-spell starts and e_{yb} the exposure.

This provides a simple exploratory model for the within-summer timing of dry-spell starts.

6.3 Estimated intensity

Figure 9 shows the empirical event rates by time bin, together with the fitted piecewise-constant NHPP intensity and the homogeneous Poisson benchmark.

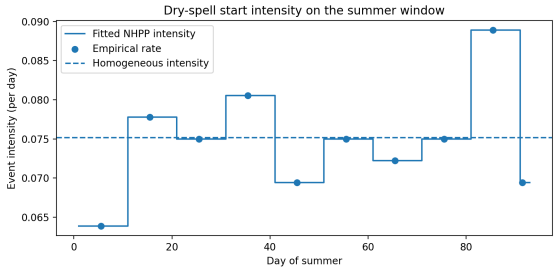


Figure 9: Empirical binwise event rates, fitted piecewise-constant NHPP intensity, and homogeneous Poisson benchmark for summer dry-spell starts.

The estimated event rate is not exactly constant over the summer season, but the fitted piecewise intensities vary only moderately across bins. In that sense, there is no strong visual evidence of pronounced within-summer non-stationarity in the occurrence rate. The piecewise NHPP improves the likelihood relative to the homogeneous model, but the gain remains limited, so the main message of this figure is that the arrival rate appears at most mildly seasonal.

This visual stability, however, is not sufficient to validate the Poisson-process assumption. For that purpose, a dedicated event-time diagnostic is required.

6.4 Model diagnostics

To assess the adequacy of the fitted point-process model, we apply a time-rescaling diagnostic based on the estimated cumulative intensity.

Under a correctly specified NHPP, the transformed inter-event times should follow an exponential distribution, and the corresponding transformed variables should be approximately Uniform$(0, 1)$.

Figure 10 shows the histogram of the transformed waits and the associated uniform Q–Q plot.

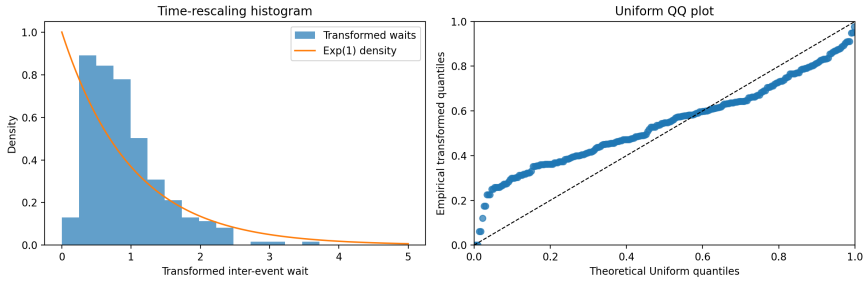


Figure 10: Time-rescaling diagnostics for the fitted dry-spell NHPP: histogram of transformed waits and uniform Q–Q plot.

The diagnostics suggest that the fitted NHPP does not adequately capture the event-time structure. In particular, the uniform Q–Q plot shows clear deviations from the diagonal and the Kolmogorov–Smirnov test yields a very small p-value.

Therefore, the NHPP is retained only as a simple exploratory benchmark rather than a fully adequate stochastic model.

6.5 Why alternative point-process objects were not retained

Two other event processes were considered in principle: the occurrence times of declustered heat-event peaks and the occurrence times of compound hot-dry events. These were not retained as the main point-process object for the following reasons.

First, heat-event peaks above a high threshold are much rarer than dry-spell starts in this dataset. Their timing is highly sensitive to threshold choice and declustering assumptions, and the resulting sample is too small to support a stable occurrence-time analysis. In our exploratory checks, Poisson-type point-process diagnostics for heat-event peaks were in fact even less satisfactory than for dry-spell starts.

Second, compound hot-dry events form an even more selective intersection process. While scientifically interesting, they are too sparse to support a separate seasonal point-process model with meaningful diagnostics.

For these reasons, dry-spell starts provide the most workable compromise: they remain meteorologically interpretable, event-based, and sufficiently frequent for exploratory occurrence-time modeling, even though the fitted NHPP ultimately proves to be inadequate in diagnostic terms.

6.6 Mark analysis

In addition to occurrence times, each dry-spell event is associated with marks describing its characteristics. Here we consider two marks: spell duration and mean temperature during the spell.

Duration mark. Let D_i denote the duration of the i-th dry spell. We model its conditional mean as a quadratic function of centered summer time:

$$\mathbb{E}[D_i \mid t_i] = \exp\big(\beta_0 + \beta_1 t_i^{(c)} + \beta_2 (t_i^{(c)})^2\big).$$

The fitted effect is weak, suggesting limited seasonal variation in spell duration.

Temperature mark. Let T_i denote the mean temperature during the i-th dry spell. We model this mark as

$$T_i = \beta_0 + \beta_1 t_i^{(c)} + \beta_2 (t_i^{(c)})^2 + \varepsilon_i.$$

In contrast, the temperature mark exhibits a clearer seasonal pattern, with higher values around mid-summer.

6.7 Interpretation

The point-process analysis was intended to complement the preceding EVT analysis by focusing on the timing of drought-related events rather than on the marginal distribution of temperature extremes.

Among the candidate event definitions, summer dry-spell starts provide the most useful occurrence-time object in this dataset, because they are clearer and substantially less sparse than heat-event peaks or compound hot-dry events. However, the time-rescaling diagnostics show that even for this comparatively favorable event definition, the fitted piecewise NHPP does not adequately describe the event-time dynamics.

Accordingly, the point-process analysis should be interpreted as exploratory. Its main contribution is not to establish a validated Poisson model for dry-spell occurrence, but rather to show that the timing of these events cannot be fully explained by a simple Poisson-type mechanism with only mild seasonal variation in intensity.

By contrast, the marked analysis remains informative: while the occurrence rate itself is not well captured by the NHPP specification, the event marks, especially mean temperature, display a clearer and more interpretable seasonal structure.